Ying TU

Department of Geography, Sustainability, Community, and Urban Studies
University of Connecticut
Address: 423 Philip E. Austin Building, 215 Glenbrook Rd, Storrs, CT 06269, USA
Email: ying.tu@uconn.edu, thutyecology@gmail.com
Website: https://thutyecology.github.io

EDUCATION

August 2019 – April 2024 — Tsinghua University
Ph.D. in Ecology

August 2015 – June 2019 — Sun Yat-sen University
B.S. in Geographic Information Science

APPOINTMENTS

May 2026 – present — University of Connecticut
Postdoctoral Research Associate

May 2024 – May 2026 — Cornell University
Postdoctoral Associate

August 2022 – August 2023 — The University of Hong Kong
Research Assistant

RESEARCH INTERESTS

- GIScience, GeoAI, and Spatial Data Science
- Remote Sensing and Environmnetal Change
- Human–Environment Interactions
- Infrastructure Access, Spatial Inequality, and Human Well-being
- Climate Adaptation, Sustainability, and Community Resilience

PUBLICATIONS

(*corresponding author, #co-first author)

Peer-reviewed articles (Google Scholar citations = 2,404)

1. **Ying Tu**, Bin Chen*, Chuan Liao*, Shengbiao Wu, Jiafu An, Chen Lin, Peng Gong, Bin Chen, Hong Wei, Bing Xu*. "Inequality in infrastructure access and its association with health disparities." *Nature Human Behaviour* 9 (2025). **[Featured on August 2025 cover]** and **[Highlighted by Tsinghua Research]**

2. **Ying Tu**, Radine Rafols, Yangyang Xu, Natalia Butler, Linah Ababneh, Feng Tao, Veerabhadran Ramanathan, Benjamin Houlton, Chuan Liao*. "Scaling up enhanced rock weathering for equitable climate change mitigation." *Communications Sustainability*. 1 (2026). **[Featured on Cornell Chronicle]**
3. **Ying Tu**, Shengbiao Wu, Bin Chen, Qihao Weng, Yuqi Bai, Jun Yang, Le Yu, Bing Xu*. "A 30 m annual cropland dataset of China from 1986 to 2021." *Earth System Science Data* 16 (2024): 2297–2316. **[ESI Highly Cited Paper]**
4. Bin Chen[#], **Ying Tu**[#], Jiafu An[#], Shengbiao Wu, Chen Lin*, Peng Gong*. "Quantification of losses in agriculture production in eastern Ukraine due to the Russia-Ukraine war." *Communications Earth & Environment* 5.1 (2024): 336.
5. **Ying Tu**, Bin Chen*, Le Yu, Yimeng Song, Shengbiao Wu, Miao Li, Hong Wei, Tingting Chen, Wei Lang, Peng Gong, Bing Xu*. "Raveling the nexus between urban expansion and cropland loss in China." *Landscape Ecology* (2023): 1-16.
6. **Ying Tu**, Bin Chen*, Jun Yang, Bing Xu*. "Olympic effects on reshaping urban greenspace of host cities." *Landscape and Urban Planning* 230 (2023).
7. **Ying Tu**, Bin Chen, Le Yu, Qinchuan Xin, Peng Gong, Bing Xu*. "How does urban expansion interact with cropland loss? A comparison of 14 Chinese cities from 1980 to 2015." *Landscape Ecology* (2021): 1-21.
8. **Ying Tu**, Bin Chen, Wei Lang, Tingting Chen, Miao Li, Tao Zhang, Bing Xu*. "Uncovering the Nature of Urban Land Use Composition Using Multi-source Open Big Data with Ensemble Learning." *Remote Sensing* 13 (2021): 4241.
9. **Ying Tu**, Wei Lang, Le Yu, Ying Li, Junhao Jiang, Yawen Qin, Jiemin Wu, Tingting Chen, Bing Xu*. "Improved Mapping Results of 10 m Resolution Land Cover Classification in Guangdong, China Using Multisource Remote Sensing Data With Google Earth Engine." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 13 (2020): 5384-5397.
10. **Ying Tu**, Bin Chen, Tao Zhang, Bing Xu*. "Regional Mapping of Essential Urban Land Use Categories in China: A Segmentation-Based Approach." *Remote Sensing* 12, no. 7 (2020): 1058.
11. **Ying Tu**, Hanlin Zhou, Wei Lang, Tingting Chen, Xun Li*, Bing Xu*. "A novel cross-sensor calibration method to generate a consistent night-time lights time series dataset." *International Journal of Remote Sensing* 41, no. 14 (2020).
12. Chuan Liao*, Shuai Zhou, Seongmin Shin, Danning Lu, Yujin Lee, Shuai Xu, **Ying Tu**, Krisztina Mosdossy, Lucinda Li, Rebecca Nelson, Johannes Lehmann. "Towards global nutrient justice via a circular bionutrient economy." *Nature Reviews Earth & Environment* (2026).
13. Yixu Zeng, Langong Hou, **Ying Tu**, Xin Chen, Jiyao Zhao, Sheng Xu, Hanguang Yu, Yi Zhou, Shengbiao Wu, Tao Liu*. "Optimizing Summer Heat Mitigation and Equity in Urban Residential Communities: Threshold-Based Evidence from 4,678 Communities in Chengdu." *Sustainable Cities and Society* (2026): 107752.
14. Ding Fei*, **Ying Tu**, Hantao Sun, and Jinpeng Yang. "Industrial parks development in Ethiopia: The institutional, spatial and transnational politics of industrial urbanism." *Cities* 176 (2026): 107156.
15. Chuan Liao, Jenny Kao-Kniffin, Matthew C Reid, Wendong Zhang, Precious Tshabalala, Erik Andrus, Zachary Butler-Jones, Jordan LaPoint, Shuai Zhou, Radine Rafols, Seongmin Shin, Mai Ichihara, Autumn Pereira, Natalia Butler, **Ying Tu**, Susan R McCouch*. "Rice Farming for Climate Change Adaptation in

the Northeastern United States." *Proceedings of the National Academy of Sciences of the United States of America* 122, no. 49 (2025).

16. Hong Wei, Bin Chen, Yi Yin, Tao Zhang, **Ying Tu**, Bing Xu*. "Distinct dominant factors of spatially heterogenous urban tree cooling identified in cities across China." *Ecological Indicators* 179 (2025): 114204.
17. Tao Liu, Le Yu*, Xiaoxuan Liu, Dailiang Peng, Xin Chen, Zhenrong Du, **Ying Tu**, Hui Wu, Qiang Zhao. "A Global Review of Monitoring Cropland Abandonment using Remote Sensing methodology: Tempo-spatial Patterns, Causes, Ecological Effects, and Future Prospects." *Journal of Remote Sensing* 5 (2025): 0584.
18. Tian Tian, Le Yu, **Ying Tu**, Bin Chen, Peng Gong*. "Mapping the Time-Series of Essential Urban Land Use Categories in China: A Multi-Source Data Integration Approach." *Remote Sensing* 16.17 (2024): 3125.
19. Hong Wei, Bin Chen, Kangning Huang, Meng Gao, Bin Fan, Tao Zhang, **Ying Tu**, Bing Xu*. "Moderating AC Usage Can Reduce Thermal Disparity between Indoor and Outdoor Environments." *Environmental Science & Technology* (2024): 10524.
20. Xinjian Wen, Fuying Yang, Jiangping Chen, **Ying Tu**, Haiyun Wang, Zhanpeng Chen, Ting Dong, Gang Xu*. "Spatial Patterns of Urban Expansion and Cropland Loss during 2017-2022 in Guangdong, China." *Heliyon* 10, no. 9 (2024).
21. Hong Wei, Yimin Chen, Bin Chen, Shuai Shi, **Ying Tu**, Bing Xu*. "Built environments, communities, and housing price: A data-model integration approach." *Applied Geography* 166 (2024): 103270.
22. Bin Chen*, **Ying Tu**, Shengbiao Wu, Yimeng Song, Yufang Jin, Chris Webster, Bing Xu, Peng Gong. "Beyond green environments: Multi-scale difference in human exposure to greenspace in China." *Environment International* 166 (2022).
23. Le Yu*, Zhenrong Du, Runmin Dong, Juepeng Zheng, **Ying Tu**, Xin Chen, Pengyu Hao, Bo Zhong, Dailiang Peng, Jiyao Zhao, Xiyu Li, Jianyu Yang, Haohuan Fu, Guangwen Yang, Peng Gong. "FROM-GLC Plus: toward near real-time and multi-resolution land cover mapping." *GIScience & Remote Sensing* 59, no. 1 (2022): 1026-1047.
24. Miao Li, Tao Zhang, **Ying Tu**, Zhehao Ren, Bing Xu*. "Monitoring Post-Flood Recovery of Croplands Using the Integrated Sentinel-1/2 Imagery in the Yangtze-Huai River Basin." *Remote Sensing* 14, no. 3 (2022): 690.
25. Bin Chen*, **Ying Tu**, Yimeng Song, David M. Theobald, Tao Zhang, Zhehao Ren, Xuecao Li, Jun Yang, Jie Wang, Xi Wang, Peng Gong, Yuqi Bai*, Bing Xu*. "Mapping essential urban land use categories with open big data: Results for five metropolitan areas in the United States of America." *ISPRS Journal of Photogrammetry and Remote Sensing* 178 (2021): 203-218.
26. Tingting Chen, Eddie CM Hui, **Ying Tu**, Wei Lang*. "Growth or Shrinkage: Discovering Development Patterns and Planning Strategies for Cross-Border Areas in China." *Journal of Urban Planning and Development* 147, no. 4 (2021): 05021046.
27. Xinyu Dou, Cuijuan Liao, Hengqi Wang, Ying Huang, **Ying Tu**, Xiaomeng Huang, Yiran Peng, Biqing Zhu, Jianguang Tan, Zhu Deng, Nana Wu, Taochun Sun, Piyu Ke, Zhu Liu*. "Estimates of daily ground-level NO2 concentrations in China based on Random Forest model integrated K-means." *Advances in Applied*

Energy 2 (2021): 100017.

28. Wenjia Cai*, Chi Zhang, Hoi Ping Suen, …, **Ying Tu** ..., Peng Gong*. "The 2020 China report of the Lancet Countdown on health and climate change." *The Lancet Public Health* (2020).
29. Ruiyun Li, Bin Chen, Tao Zhang, …, **Ying Tu** ..., Bing Xu*. "Global COVID-19 pandemic demands joint interventions for the suppression of future waves." *Proceedings of the National Academy of Sciences* 117, no. 42 (2020): 26151-26157.
30. Peng Gong*, Bin Chen, Xuecao Li, …, **Ying Tu** ..., Bing Xu*. "Mapping essential urban land use categories in China (EULUC-China): preliminary results for 2018." *Science Bulletin* 65, no. 3 (2020): 182-187.

Conference proceedings

1. **Ying Tu**, Bin Chen, Le Yu, Qinchuan Xin, Peng Gong, and Bing Xu*. "Urban-Expansion Driven Farmland Loss Follows with the Environmental Kuznets Curve Hypothesis: Evidence from Temporal Analysis in Beijing, China." *In Geoinformatics in Sustainable Ecosystem and Society*, pp. 394-412. Springer, Singapore, 2019.
2. Shengjie Liu, Haowen Luo, **Ying Tu**, Zhi He, and Jun Li*. "Wide contextual residual network with active learning for remote sensing image classification." *In IGARSS 2018-2018 IEEE International Geoscience and Remote Sensing Symposium*, pp. 7145-7148. IEEE, 2018.

TEACHING

Teaching Experience

FA 2026	University of Connecticut	GSCU 3512 Introduction to Spatial Data Science	Guest Lecturer & Lab Module Developer
FA 2026	University of Connecticut	GSCU 1500 Introduction to Digital Earth	Guest Lecturer
SP 2026, 2025	University of Connecticut	GEOG 4095/5895 Introduction to Geospatial Artificial Intelligence	Guest Lecturer
FA 2025	Kennesaw State University	GEOG 1112K Introduction to Weather and Climate	Teaching Demonstration
SP 2025	Cornell University	GDEV 3170/5170 Land Systems and Sustainability	Guest Lecturer & Lab Module Developer
FA 2022	The University of Hong Kong	LAND 7176 GIS for Landscape Architecture	Teaching Assistant
SP 2021	Tsinghua University	The Way of Academia	Teaching Assistant
FA 2020	Tsinghua University	Global Change and Sustainable Development	Teaching Assistant

Student Mentoring

Graduate Students

Time	Institution	Name	Level	Contributions and Outcomes
2026–	University of Connecticut	Mr. Z. Chen	Ph.D.	Data collection and analysis, research design, dissertation support, and grant proposal development
2026–	University of Connecticut	Ms. J. Huang	Ph.D.	Data collection and analysis, research design, dissertation support, and grant proposal development
2026–	University of Connecticut	Ms. J. Zhao	Ph.D.	Joint seed grant proposal development
2026–	University of Connecticut	Ms. H. Shi	S.J.D.	Research design and dissertation support
2024–2026	Cornell University	Ms. N. Butler	Ph.D.	Data processing and research support
2025–2026	Cornell University	Mr. Z. He	Ph.D.	Data collection and analysis, research design, and dissertation support
2025–2026	Cornell University	Ms. Y. Pan	Master's	Data collection and analysis, capstone project guidance, and career advising (now a facility engineer at Headway Technologies)
2022–2023	The University of Hong Kong	Ms. Q. Qin	Master's	Research design and thesis support
2022–2023	The University of Hong Kong	Mr. Y. Yuan	Master's	Research design and thesis support

Undergraduate Students

Time	Institution	Name	Contributions and Outcomes
2026–	University of Connecticut	Ms. Y. Zhao	Research support and graduate application advising
2025–2026	Cornell University	Ms. Y. Wang	Research collaboration and manuscript coauthorship
SP 2021	Tsinghua University	Mr. Y. Dai	Coursework support and graduate application advising (now a Ph.D. student at MIT)
SP 2021	Tsinghua University	Mr. K. Chen	Coursework support and graduate application advising (now a Ph.D. student at the University of Chicago)
FA 2020	Tsinghua University	Mr. Y. Zhang	Coursework support and graduate application advising (now a Ph.D. student at the University of Michigan)
FA 2020	Tsinghua University	Mr. J. Yin	Coursework support and career advising

FUNDING

Proposals Submitted

- *Perceiving Neighborhood Safety: An AI-Enabled Study of Environmental Appraisal and Visit Intentions among Sexual Minority Men*. Institute for the Brain and Cognitive Sciences (IBACS) Seed Grant, University of Connecticut. My role: PI; Team with Dr. Hanlin Zhou, Dr. Chenglin Hong, Mr. Zhang Chen, and Ms. Jie Zhao. Amount requested: **$24,965.56**
- *Translating Human Activity Changes During Extreme Heat into Heat–Health Action Priorities in New England*. CAFE Health and Extreme Weather Actionable Solutions Grants Program Application, pilot project award, funded by NIH U24-ES035309 (2026–2027). My role: Co-PI; Team with Dr. Hanlin Zhou serving as PI and Mr. Zhang Chen. Amount requested: **$20,000**
- *Environmental Risks and Perceptual Trapping: Using Geospatial AI to Link Objective Impedance and Subjective Perceptions for Age-Friendly Mobility*. Center on Aging, Health, and Environment (CACHE) Seed Grant, supported by the National Institute on Aging (NIA) "Aging and Health in a Changing Climate" R61AG086854 (2026–2027). My role: Co-PI; Team with Dr. Hanlin Zhou serving as PI and Ms. Jing Huang. Amount requested: **$25,000**

Funded Projects

- *Sorghum for sustainability? Enabling water savings from crop switching in Kansas*. Environmental Defense Fund (2025–2027). My role: Participant.
- *SCC-CIVIC-PG Track A: Community-driven Innovations in Rice for Climate Adaptation in New York State (CIRCA-NYS)*. National Science Foundation (NSF) (2024–2027). My role: Participant.
- *Agricultural Floodplain Rice Farming as Adaptation to Climate Change in New York State*. United States Department of Agriculture (USDA), National Institute of Food and Agriculture (2024–2025). My role: Participant.
- *Agricultural Floodplain Rice Farming as Adaptation to Climate Change in NYS: An RCT Approach*. Towards Sustainability Foundation (2024). My role: Participant.
- *Agricultural Floodplain Rice Farming for Just Technological Futures: An Environmental Justice Perspective*. Cornell CALS Moonshot Seed Grant (2024). My role: Participant.
- *From 2D to 3D greenspace exposure: A novel "Pixel-Voxel" integration of green volume estimates and environmental exposure models.* Research Grants Council (RGC) of Hong Kong General Research Fund (GRF) (2024-2026). My role: Participant.
- *Parcel-based urban land use classification using domain adaptations of sample, feature, and model*. National Natural Science Foundation of China (NSFC) Young Scientists Fund (2023–2025). My role: Participant.
- *Green and low-carbon development monitoring and diagnosis optimization application in megacities*. National Key Research and Development Program of China (2022–2026). My role: Participant.
- *Integrated Assessment of Urban Climate-related Risks: Hazard, Exposure, and Vulnerability*. URC Seed Fund for Strategic Interdisciplinary Research Scheme,

The University of Hong Kong (2022-2025). My role: Participant.

- *Global change impact on human infectious diseases and health, and its urban responses and sustainable development*. Open Research Program of the International Research Center of Big Data for Sustainable Development Goals (2022–2023). My role: Participant.

PRESENTATIONS & TALKS

1. **Ying Tu**. "Agricultural Sustainability in Coupled Human–Environment Systems: A Remote Sensing and GeoAI Perspective." *Invited Talk at Zhejiang University*, Hangzhou, Zhejiang, China. December 2025.
2. **Ying Tu**, Chuan Liao. "Crop Switching as a Key Strategy for Climate Change Adaptation: Insights from Floodplain Rice Farming in the United States." *AGU Annual Meeting*, New Orleans, LA, USA. December 2025.
3. **Ying Tu**. "GeoAI for Coupled Human–Environment Systems: Environmental Monitoring, Equity, and Sustainability." *Invited Talk at Kennesaw State University*, Atlanta, GA, USA. December 2025.
4. **Ying Tu**, Ding Fei. "Urbanizing through Special Economic Zones? Territorial Outcomes of Chinese-Contracted Industrial Parks in Ethiopia." *AAG Annual Meeting*, Detroit, MI, USA. March 2025.
5. **Ying Tu**, Chuan Liao, Jenny Kao-Kniffin, Matthew Reid. "Assessing the Suitability of Agricultural Floodplain Rice Farming as Adaptation to Climate Change in the United States." *AGU Annual Meeting*, Washington, D.C, USA. December 2024.
6. **Ying Tu**, Chuan Liao. "Climate Adaptation and the Potential for Flood-Tolerant Agriculture in New York State." *Cornell Rice Field Day*, Ithaca, NY, USA. October 2024.
7. **Ying Tu**, Chuan Liao. "Remote Sensing and Climate Adaptation in Agricultural Floodplain Regions." *Cornell Cooperative Extension (CCE) Executive Leadership Conference*, Freeville, NY, USA. September 2024.
8. **Ying Tu**, Bin Chen, Jun Yang, Bing Xu. "Olympic effects on reshaping urban greenspace of host cities." *The 3rd International Forum on Big Data for Sustainable Development Goals*, Beijing, China. September 2023.
9. **Ying Tu**, Bin Chen, Bing Xu. "Raveling the nexus between urban expansion and cropland loss in China." *AAG Annual Meeting* (Virtual). February 2022.
10. **Ying Tu**, Bin Chen, Le Yu, Qinchuan Xin, Peng Gong, Bing Xu. "Urban-Expansion Driven Farmland Loss Follows with the Environmental Kuznets Curve Hypothesis: Evidence from Temporal Analysis in Beijing, China." *The 7th International Conference on Geoinformatics in Sustainable Ecosystem and Society*, Guangzhou, China. November 2019.

AWARDS

2024	Outstanding Doctoral Dissertation of Tsinghua University
2023	First Scholarship of Tsinghua University
2022	National Scholarship of China (for Doctoral Students)

2021	Outstanding Teaching Assistant of Tsinghua University
2021	Second Scholarship of Tsinghua University
2020	First Scholarship of Tsinghua University
2020	Best Report in the 610th Doctoral Forum of Tsinghua University
2019	Best Student Paper in the Geoinformatics in Sustainable Ecosystem and Society (GSES) Conference
2019	Outstanding Undergraduate Thesis of Sun Yat-sen University
2019	"Top 10 Graduation Student" of School of Geography and Planning, Sun Yat-sen University
2017	National Scholarship of China (for Undergraduate Students)
2016	First Scholarship of Sun Yat-sen University
2015	First Scholarship of Sun Yat-sen University

PROFESSIONAL SERVICE

Editorial Service

Topic Editor for "Advances in Urban Resilience for Sustainable Futures", including the journals *Buildings*, *Land*, *Sustainability*, *Urban Science*, and *Water*.
Co-edited with Dr. Hanlin Zhou, Dr. Wei Lang, and Dr. Tingting Chen.

Guest Editor of the Special Issue "Sustainable Application of Land Surveying and Mapping in Land Resource Development and Management" in *Sustainability*.

Conference Service

Session Chair for "Towards Healthy and Resilient Communities: Advancing Environmental Health Research with GeoAI and Human-Centered Geospatial Data", American Association of Geographers (AAG) Annual Meeting (2027)

University Service

Invited Panelist, Postdoc Appreciation Week, University of Connecticut (2026)

Departmental Service

GSCU Colloquium Committee Member, Department of Geography, Sustainability, Community, and Urban Studies, University of Connecticut (2026–2027)

Grant Review

Taylor Geospatial Institute Geospatial Innovation for Food Security Challenge (2025)

Journal Review (Total = 52)

Cities (2026x2)
Communications Earth & Environment (2025)
Computers, Environment and Urban Systems (2025)
Earth's Future (2025x2)
Earth System Science Data (2024)

Ecological Indicators (2026x2)
Environmental Research (2026)
Environmental Science & Technology (2024)
Food Security (2024)
Geo-spatial Information Science (2023)
Geography and Sustainability (2026)
GIScience & Remote Sensing (2023, 2025)
Habitat International (2026x3)
Infrastructure (2024)
International Journal of Agricultural Sustainability (2024)
International Journal of Remote Sensing (2021x3, 2022)
International Journal of Digital Earth (2023x2, 2024)
International Journal of Electrical and Computer Engineering (2021)
International Journal of Environmental Research and Public Health (2023)
ISPRS Journal of Photogrammetry and Remote Sensing (2023, 2025)
International Journal of Applied Earth Observation and Geoinformation (2026)
Journal of Environmental Management (2025)
Land (2022x2, 2023)
Land Use Policy (2024, 2026)
Landscape and Urban Planning (2026x2)
Next Sustainability (2026)
Regional Environmental Change (2024)
Remote Sensing (2023x3)
Remote Sensing in Ecology and Conservation (2022)
Remote Sensing of Environment (2024)
Resources, Environment and Sustainability (2026)
Scientific Data (2024)
Sustainable Horizons (2026)
Transportation Research Interdisciplinary Perspectives (2026)

Academic Affiliations

American Association of Geographers (AAG)
American Geophysical Union (AGU)

LANGUAGES

Chinese (native); English (professional, IELTS 7.5); Cantonese (fluent)

SKILLS

Programming: Python, R, JavaScript, Matlab, C/C++, VB, etc.

Tools: ArcGIS, ArcGIS Pro, QGIS, Google Earth Engine, Microsoft Planetary Computer, ENVI, etc.